

Общая физика

Земная физика

$$x(t), \quad v(t) = \frac{dx}{dt} \quad a(t) = \frac{dv}{dt}$$

$\xrightarrow{\text{групп.}}$ $\xrightarrow{\text{групп.}}$
 $\xleftarrow{\text{инт.}}$ $\xleftarrow{\text{инт.}}$

① Дано:

$$v_0 = 30 \text{ км/ч}$$

$$a_0 = -4 \text{ м/с}^2$$

Найти:

Решение:

$$1) \frac{dv}{dt} = -a_0 \Rightarrow v(t) = a_0 t + C$$

$$2) v(0) = v_0$$

$$a_0 \cdot 0 + C = v_0 \Rightarrow C = v_0 \Rightarrow$$

$$\Rightarrow v(t) = a_0 t + v_0$$

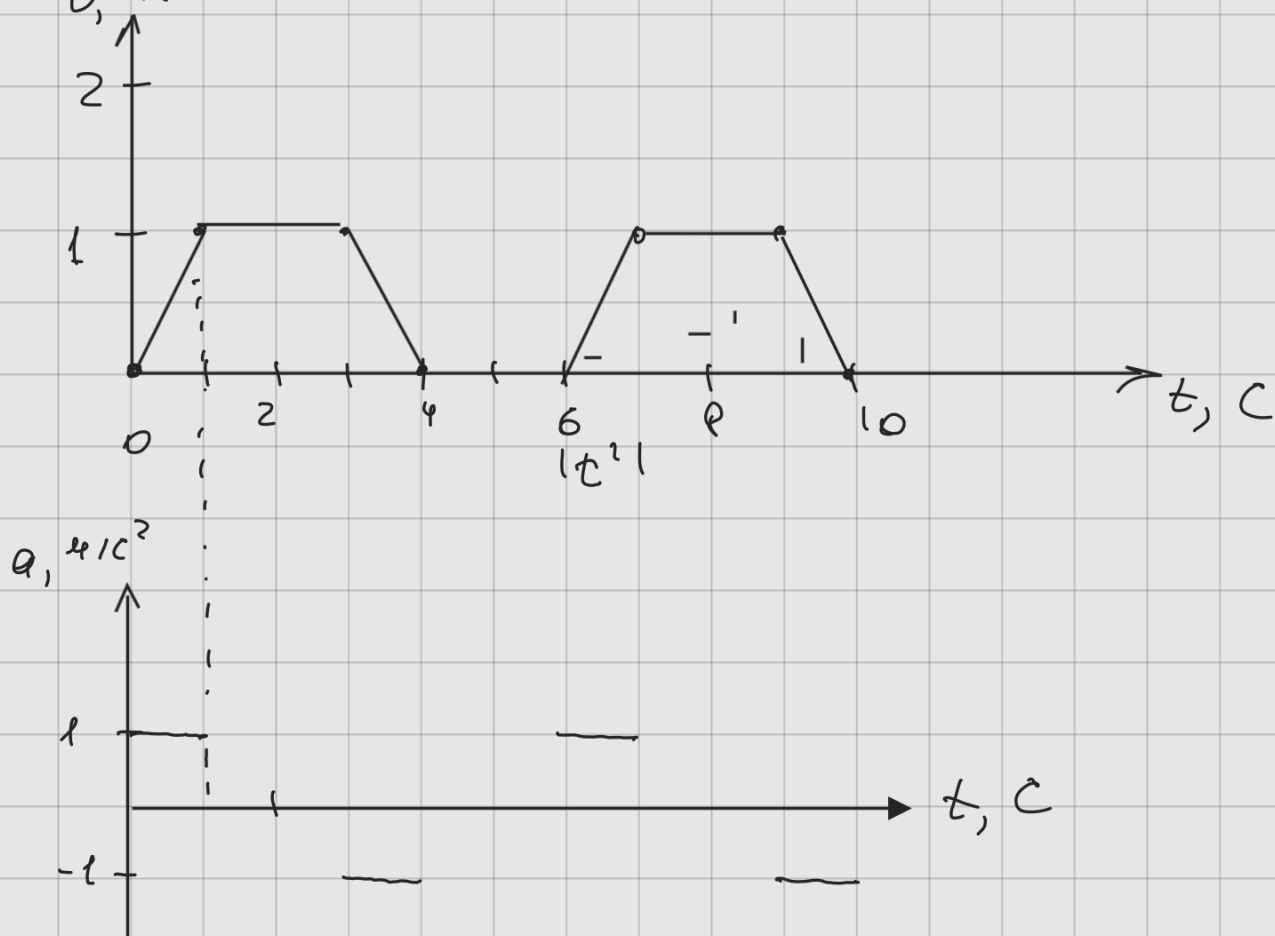
$$3) \int dx = \int v dt = \int (a_0 t + v_0) dt$$

$$\Delta x = \frac{a_0 t^2}{2} \Big|_0^{t_{\text{ост}}} + v_0 t \Big|_0^{t_{\text{ост}}}$$

$$v(t_{\text{ост}}) = 0 \Rightarrow t_{\text{ост}} = -\frac{v_0}{a_0} \Rightarrow$$

$$\frac{a_0}{2} \left(-\frac{v_0}{a_0}\right)^2 + v_0 \left(-\frac{v_0}{a_0}\right) = \frac{v_0^2}{2a_0} - \frac{v_0^2}{a_0} = \frac{-v_0^2}{2a_0} = \frac{-\left(\frac{30}{3,6}\right)^2}{2 \cdot (-4)}$$

$$\approx 8,7 \text{ м}$$



$$\underline{t \in (0, 1)}: \\ x(t) = \int_0^t t dt = \frac{t^2}{2} = \int_0^t v(t) dt = \\ x(1) = \frac{1}{2}$$

$$\underline{t \in (1, 3]}: \Rightarrow x(t) = \frac{1}{2} + \int_1^t 1 dt = \frac{1}{2} + t - 1 = t - \frac{1}{2} \\ x(3) = \frac{5}{2}$$

$$\underline{t \in (3, 6]}: \Rightarrow x(t) = \frac{5}{2} + \int_3^t (4-t) dt = \frac{5}{2} + 4(t-3) - \frac{t^2}{2} + \frac{9}{2} = \\ = 7 + 4t - 12 - \frac{t^2}{2} = 4t - \frac{t^2}{2} - 5$$

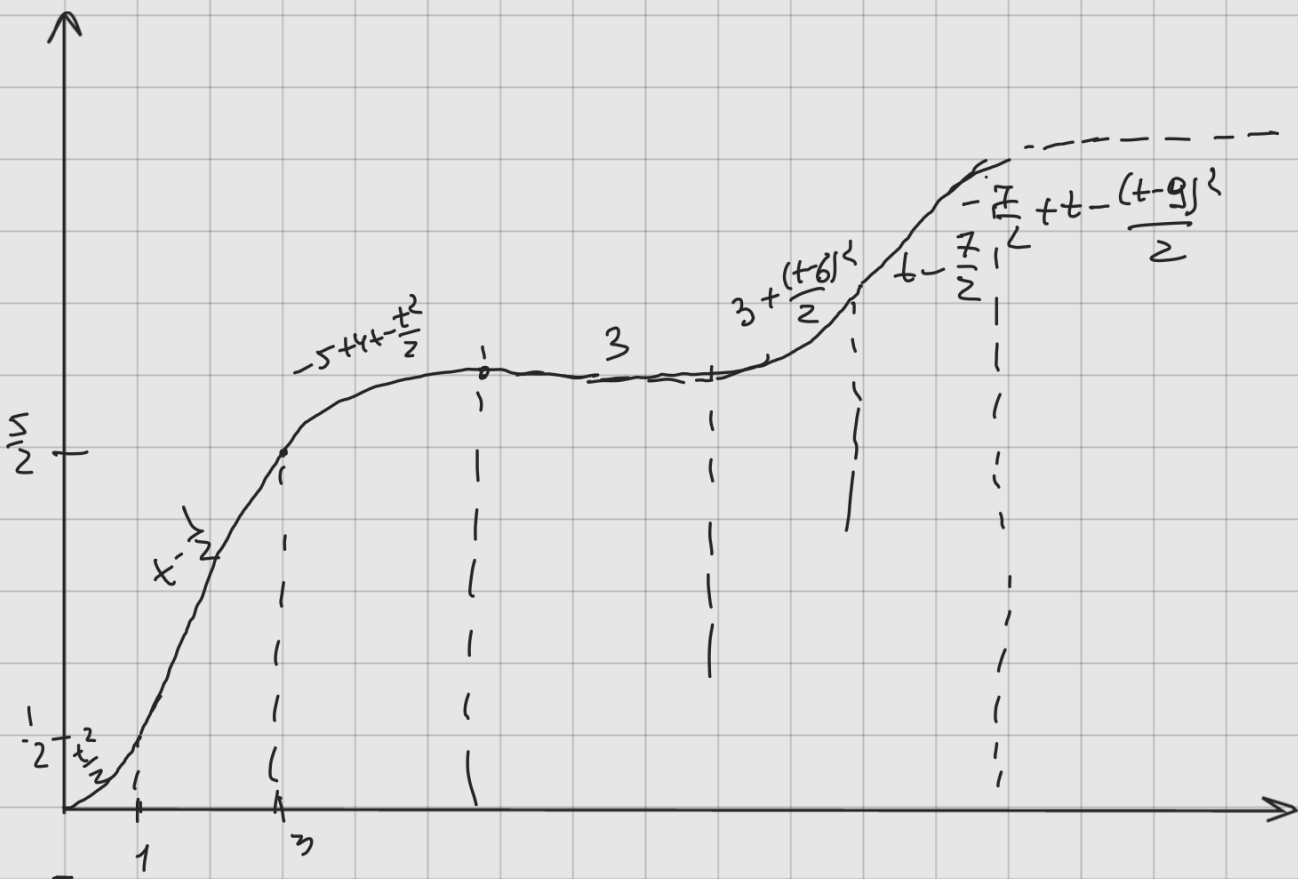
$$\underline{(5, 6]} \\ x(t) = 3 + \int_0^{t'} t' dt' = 3 + \frac{t'^2}{2} = 3 + \frac{(t-6)^2}{2}$$

$$t \in (7, 9]: x(t) = \frac{7}{2} + \int_0^t 1 dt'' = \frac{7}{2} + t'' = \frac{7}{2} + (t - 7)$$

$$= t - \frac{7}{2}$$

$$t \in (9, 10]: x(t) = \frac{11}{2} + \int_0^t (1 - t''') dt''' = \frac{11}{2} + t''' - \frac{t'''^2}{2} =$$

$$= t - \frac{7}{2} - \frac{(t - 9)^2}{2}$$



Дано:

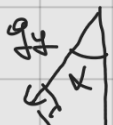
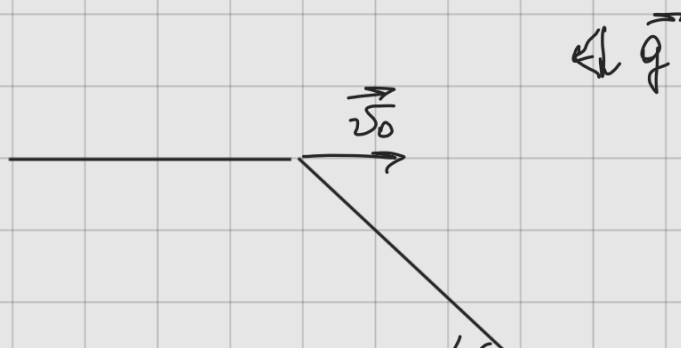
$$\alpha = 45^\circ$$

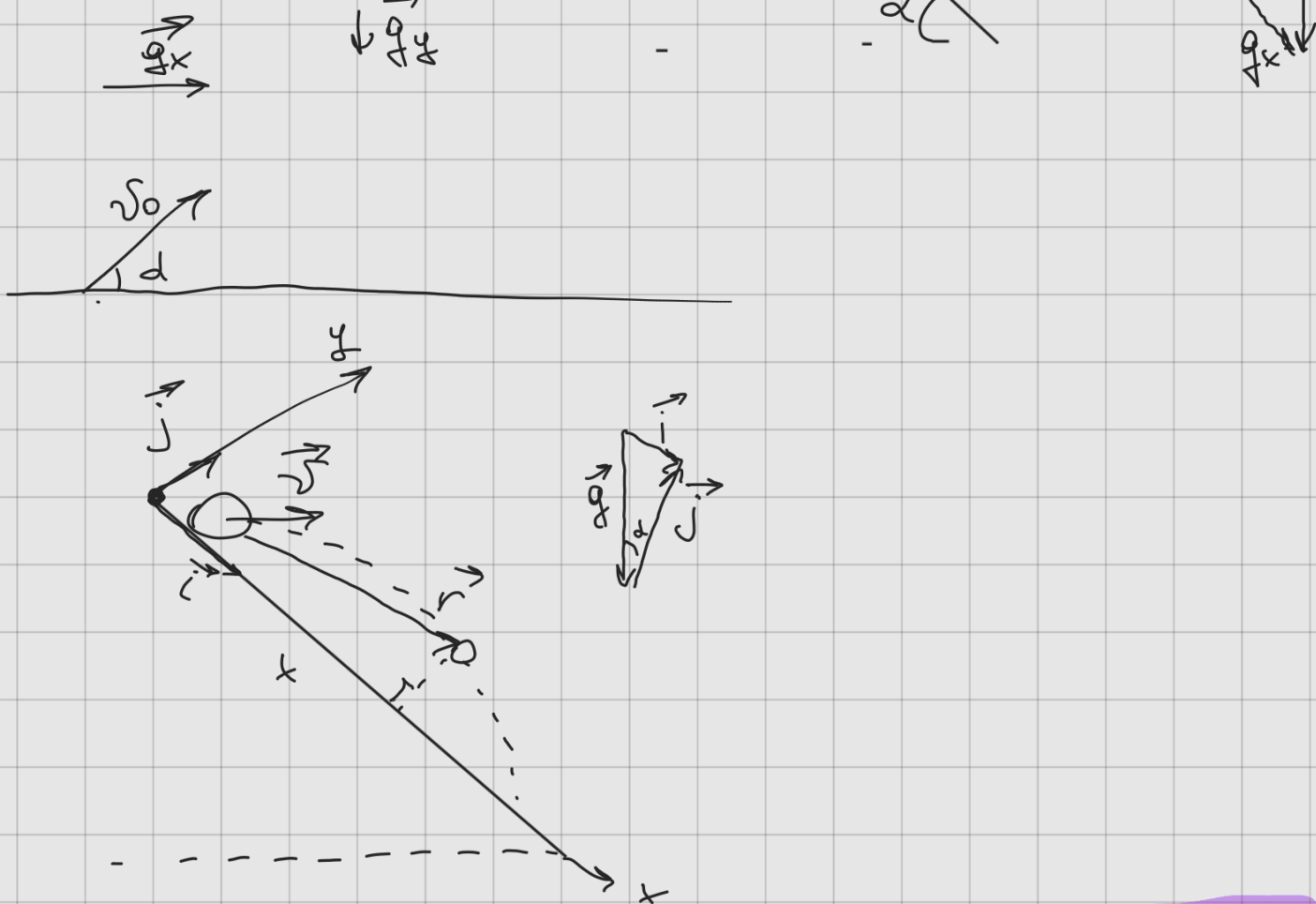
$$v_0 = 10 \text{ м/с}$$

$$g = 9,8 \text{ м/с}^2$$

Найти: ℓ , v_k

Решение:





$$\frac{d\vec{v}}{dt} = \vec{g} \Rightarrow \vec{v}(t) = \vec{g}t + \vec{v}_0$$

$$\vec{r}(t) = \frac{\vec{g}}{2}t^2 + \vec{v}_0t + \vec{r}_0$$

$$x = \vec{r} \cdot \vec{i} = |\vec{r}| \cdot |\vec{i}| \cdot \cos \alpha = \vec{g} \cdot \vec{i} \cdot \frac{t^2}{2} + \vec{v}_0 \cdot \vec{i} t + \vec{r}_0 \cdot \vec{i}$$

$$= g \sin \alpha \cdot \frac{t^2}{2} + v_0 \cos \alpha t$$

$$y = \vec{r} \cdot \vec{j} = -g \cos \alpha \frac{t^2}{2} + v_0 \sin \alpha t$$

$$y(0) = 0 \Rightarrow (t=0) \vee (t_n = \frac{2 v_0 \sin \alpha}{g \cos \alpha})$$

$$x(t_n) = \left(\frac{2 v_0^2 \sin^3 \alpha}{g \cos^2 \alpha} + \sin \alpha \right) = \frac{2 v_0^2 \sin \alpha}{g \cos^2 \alpha}$$

gloss &

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